

```

void display() {
    cout << "Name: " << name << "Marks: " << marks << endl;
}

int main() {
    int n = 5;
    Student *ptr = new Student [n]; // dynamic array of object
    // input
    for (int i = 0; i < n; i++) {
        ptr[i].input();
    }

    // output
    for (int i = 0; i < n; i++) {
        ptr[i].display();
    }

    delete [] ptr; // free memory.
    return 0;
}

```

how memory works

```

ptr → [obj 1] [obj 2] [obj 3]
      ↑       ↑       ↑
      ptr    ptr+1   ptr+2

```

Accessing object
Method 1 (Easy)

ptr[i].name
Method 2 (Pointer style)

Day 10 (Array of Object using Pointers)

Q What is Array of Object?

Normally: `Student S;` // creates only 1 object

`Student S[3];` // creates 3 objects

Using Pointers (Dynamic way)

Instead of fixed size, we use pointers

`Student *Ptr = new Student[3];`

Ynis means

- memory is created in heap
- 3 objects are created dynamically.

Example

```
#include <iostream>
```

```
using namespace std;
```

```
class Student {
```

```
public:
```

```
    string name;
```

```
    int marks;
```

```
    void input();
```

```
    cout << "Enter name and marks  
    name >> marks;
```



```
Student *ptr = &s; // pointer to object
s.marks = 90;
ptr -> show(); // using pointer.
return 0;
```

3.

how to Access members:

two ways: s.marks (using object)
ptr -> marks (using pointer)

Arrow operator

shortcut for

$ptr \rightarrow x \equiv (*ptr).x$

Example

$(*ptr).marks = 90;$ same as $ptr \rightarrow marks = 90;$

dynamic object with pointers

```
#include <iostream>
using namespace std;
```

```
class A {
```

```
public:
```

```
void show() {
```

```
cout << "hello" << endl;
```

```
}
```

```
};
```

```
int main() {
```

```
A* obj = new A;
```

```
obj -> show();
```

```
delete obj;
```

```
return 0;
```

3.

delete ptr;
Example: delete p;

Key important?

If you don't delete memory leak.

Problem statement 1. Dynamic integer.

Problem statement 2. Dynamic array input

Problem 2: sum of dynamic array.

dynamic array

```
int *arr = new int[5]
```

```
arr[0] = 10;
```

```
arr[1] = 20;
```

```
delete[] arr;
```

Day-28 (Pointers to object) / (Arrow operator)

Pointer to object is basically the pointer that stores the address of an object.

Syntax

```
classname *ptr;
```

Example

```
#include <iostream>
using namespace std;
```

```
class Student {
```

```
public:
```

```
int marks;
```

```
void show() {
```

```
cout << "Marks: " << marks << endl;
```


$(ptr + i) \rightarrow \text{name}$. (Both are same.)

Important

Student *ptr = new student[n]

This does;

Allocate memory

call constructors for each object

(Never modify original
when using dynamic
array)

Deleting memory

delete[] ptr

Just use [] because:
• It's an array.
• Free all object

Super

Practice Problem: Student Record System.

- Requirement :- Create a class student with
• name, marks.

• In main $\rightarrow$

- take number of student
- Create dynamic array
- Take input
- Display details.
- find and display student with highest marks.